

Information Retrieval for Climate Impact

Report on the MANILA24 Workshop

Editors

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Abstract

The purpose of the MANILA24 Workshop on information retrieval for climate impact was to bring together researchers from academia, industry, governments, and NGO's to identify and discuss core research problems in information retrieval for climate impact. The workshop aimed to foster collaboration by bringing communities together that have so far not been very well connected – information retrieval, systematic reviews, and climate change. The workshop brought together a diverse set of researchers and practitioners interested in contributing to the development of a technical research agenda for information retrieval for climate impact.

1 Introduction

Human-induced climate change, including more frequent and intense extreme events, has caused widespread adverse impacts and related losses and damages to nature and people, beyond natural climate variability [IPCC, 2022a]. The Intergovernmental Panel on Climate Change (IPCC) is the leading international body for assessment of climate change. Approximately every six years, the IPCC releases a series of reports on the different aspects of climate change based on large-scale peer review of the latest scientific research. This report is divided into three reports produced by different working groups. In particular, Working Group II (WGII) of the IPCC “assesses the impacts, adaptation, and vulnerabilities related to climate change, from a world-wide to a regional view of ecosystems and biodiversity, and of humans and their diverse societies, cultures

*Affiliations not shown due to space limitations. See Appendix A for details.

and settlements. It considers their vulnerabilities and the capacities and limits of these natural and human systems to adapt to climate change and thereby reduce climate-associated risks together with options for creating a sustainable future for all through an equitable and integrated approach to mitigation and adaptation efforts at all scales” [IPCC, 2024].

Fully using the available knowledge on emerging climate change impacts is key to informing global policy processes as well as regional and local risk assessments and on-the-ground action on climate adaptation [Schleussner and Fyson, 2020]. The exponential growth in peer-reviewed scientific publications on climate change is pushing manual expert assessments to their limits [Callaghan et al., 2021, 2020; Joe et al., 2024]. While literature aggregated on the level of continents or world regions might be useful to the global policy process, informing concrete climate adaptation typically requires more highly localized and contextualized information on climate impacts [Conway et al., 2020]. Tracking the effectiveness and progress of adaptation actions has proven difficult [Sijtsma et al., 2024] – any attempt to track adaptation progress will need to be capable of rapidly handling large and varied datasets, while remaining sensitive to highly localized and contextualized information.

This workshop report brings together researchers from the information retrieval, natural language processing, and climate science communities in an attempt to develop an agenda to advance information retrieval for climate impact. We begin by dissecting the problem: what is the information need addressed by the IPCC WGII? We then switch to methodologies, and in particular systematic reviews. We review different resources available (and/or needed) to support information retrieval for climate impact, including test sets and implementations. We then address the question of how to make new technological advances in information retrieval work as part of the IPCC WGII assessment workflow. The paper concludes with a broader perspective. To remain focused, our discussion and analysis is centered around IPCC WGII and its mission – we believe, however, that many of our questions and suggestions have the potential to contribute to the workflows of other IPCC working groups and task forces.

2 The Information Need

The IPCC aims to provide governments at all levels with scientific information that they can use to develop climate policies [United Nations, 2024]. The IPCC is divided into three working groups (WGs). WGI deals with the physical science basis of climate change, WGII with climate change impacts, adaptation and vulnerability, and WGIII with mitigation of climate change. The IPCC does not conduct its own research, run models, or make measurements of climate or weather phenomena. Its role is to assess the scientific, technical and socio-economic literature relevant to understanding climate change, its impacts, future risks, and options for adaptation and mitigation. Author teams assess all such information from any source that is to be included in the report.

Approximately every six years, the IPCC releases a series of reports on the different aspects of climate change based on large-scale peer review of all the latest scientific research. In early 2022, as part of the IPCC’s sixth assessment report (AR6), the IPCC released the report of WGII on impacts, adaptation and vulnerability, which “assesses the impacts of climate change, looking at ecosystems, biodiversity and human communities at global and regional levels. It also reviews vulnerabilities and the capacities and limits of the natural world and human societies to adapt to climate change” [IPCC, 2022a]. The report consists of 18 chapters focusing on ecosystems,

sectors, sustainable development goals, and regions (and 7 cross-chapter papers), and an integrated technical summary. The AR6 WGII report pulls together evidence and findings from more than 34,000 papers; it is written by 270 authors from 67 countries.

The assessment report currently in development (AR7) faces a twin challenge: assessing and synthesizing a fast-growing amount of literature related to climate change, and addressing relevant topics and areas where little data is available in the published English scientific literature. We believe that IR can potentially help address both aspects, by (i) helping IPCC author teams sift through the literature, and (ii) exposing relevant information previously hidden from author teams due to various barriers (language, etc.).

2.1 Proposed Research

We propose an agenda driven by the ambition to produce open evidence synthesis, based on principles of transparent information gathering, traceability, and information quality. Fairness and mitigating different types of bias (geographic, cultural) are important proposed research lines. Minorities, those not in the system, need to be accounted for. Existing AI tools work well with large amounts of data, so have the potential to exacerbate imbalances in data coverage and literature coverage. How do we make sure AI can help with topics or areas where less information is available? Another line of research has to do with the data sources used for evidence synthesis: (i) (capturing and) using domain-specific information, (ii) the usage of grey literature, i.e., the diverse and heterogeneous body of material available outside, and not subject to, traditional academic peer-review processes such as social media posts, preprints, and policy documents [Adams et al., 2017], (iii) indigenous and/or local knowledge, especially if it is not available online, and (iv) (qualitative) data collected through grass roots and/or citizen data science efforts.

It is important that retrieval and analysis models can handle multi-modal data (i.e., text, numerical, images, etc.). Only published data currently goes into the IPCC review process, and this is unlikely to cover, e.g., remote sensing data. The final direction concerns important guardrails that apply to the evidence synthesis process. One is that there is zero tolerance for black boxes – this is important to avoid undermining the credibility of the synthesis with policy makers and governments. And another concerns the carbon emissions when choosing information retrieval (IR) approaches (i.e., energy requirements of using large language models (LLMs)). Do benefits of the model/data/research outweigh the carbon costs?

2.2 Research Challenges

These general research questions manifest themselves along the entire evidence synthesis chain and motivate a broad range of concrete research directions to be investigated concerning the information need that IPCC addresses. Some of these have to do with the scope of the evidence synthesis, some concern data acquisition and quality assessment, and some with contextual factors.

Data gathering and sharing. If more diverse sources of evidence are integrated into the evidence synthesis process, how can we ensure open, transparent data collection and sustainable sharing?

Data quality. A common suggestion is that evidence should include more diverse sources, especially grey literature. However, combining different datasets or manually annotating data is time-intensive, and grey literature in particular is difficult to work with [Sietsma et al., 2024]. How can local expert knowledge be embedded in the evidence synthesis process, while ensuring that data/literature imbalances are not exacerbated? How can acceptable data be created from unconventional sources?

Climate adaptation. It has been claimed that robust synthesis of climate *adaptation* literature and insights was under-represented in climate impact assessments, including the most recent version of the WGII report [Berrang-Ford et al., 2021]. How to address and capture successful adaptations, whether technological, institutional, behavioral, or nature-based [O’Neill et al., 2022]. How to avoid “maladaptations” that may increase vulnerability, lock-in, or unequal impacts [Barnett and O’Neill, 2013]? How to measure outcomes of adaptation actions and aggregate them, e.g., based on similarity of contexts, precision, causality?

Timeliness. A single IPCC assessment cycle lasts approximately six years and the resulting syntheses are difficult to keep up to date. How can more recent statistics, figures, and findings from research papers lead to “updated” reports? What would an acceptable “living evidence” in the context of the IPCC look like [Elliott et al., 2017]?

Evaluation. How can we evaluate “fit to information need” of assessment reports or of the various summaries, such as the summary for policymakers [IPCC, 2022b] or the technical summary [Pörtner et al., 2022], that are generated based on the reports?

2.3 Obstacles and Risks

To enable this research we need broad collaborations between IR researchers and climate impact researchers. Finding effective ways of working together and finding a shared vocabulary requires considerable effort that some researchers may not be able to afford or have insufficient institutional support, resources or recognition for [Allan et al., 2018]. An important risk concerns the continuation or even exacerbation of imbalances in coverage in IPCC reports by focusing on areas and topics that have large amounts of data available for training. A potential obstacle for bringing recent advances in IR technology to bear on the IPCC’s evidence synthesis workflow is the broad – and successful – uptake of technologies that are often referred to as “opaque” or “black-box” technologies such as LLMs and generative AI. Another key constraint is that the IPCC report production process cannot be changed, potentially limiting the integration of important innovations in the evidence synthesis process.

3 Methodology

So far, the IPCC has followed a conventional approach for its evidence synthesis [Cooper et al., 2018]. This means: (i) querying (which includes (a) protocol definition and (b) search strategy development), (ii) screening (which includes (a) study abstract screening and (b) study full-text

screening), and (iii) aggregating (which includes (a) study synthesis and results preparation and (b) dissemination of systematic review). IPCC's approach can be characterized using the following dimensions:

Rigorous assessment Use established methods for comprehensive reviews, aligning with IPCC's mission in climate literature.

Database utilization Employ robust academic databases such as Scopus and Web of Science for extensive literature search.

Literature screening Implement a systematic selection process to ensure the inclusion of relevant studies.

Publication coding Use qualitative analysis software to code and categorize information effectively.

Scholarly perspectives Evaluate the gathered facts and identify research gaps to provide a well-rounded scholarly insight.

Data visualization Create visual representations such as diagrams, word clouds, and tables to present the synthesized data clearly.

Connect with practitioners Summarize key findings to inform policy implications and engage with practitioners for real-world application.

IPCC's current approach to evidence synthesis has several shortcomings: (i) the growth of the literature: human-driven information collection skews search relevance, hence tool support is needed to manage and streamline new literature; (ii) information comes in diverse formats, both structured and unstructured: fragmented information hinders understanding and analysis, hence tool support is needed to integrate structured and unstructured data; and (iii) there is a significant time lag due to the six year cycle, for which *living* systematic reviews may be a way forward [Elliott et al., 2017].

3.1 Proposed Research

Developing evidence synthesis methodologies that help address shortcomings in IPCC's assessment process requires new research on a broad range of information retrieval topics subject to a number of constraints and guardrails. In particular, information retrieval methods are needed for identifying, integrating, understanding, and selecting relevant literature and information extraction methods for (i) efficiently synthesizing relevant data and (ii) improving accuracy and relevance of extracted information. The main constraints which these methods need to operate under are (i) eliminating hallucination; (ii) mitigating bias by enhancing the selection process and promoting fairness and inclusivity; and (iii) maintaining human accountability and ensure transparency in AI-assisted processes.

3.2 Research Challenges

Challenges are faced on each of the areas that the proposed research covers. The proposed research touches on the core of evidence synthesis: the way of working, the data, data enrichment, the synthesis and summarization process, its timeliness, and the role of people in the process.

Protocol development How should evidence synthesis protocols be re-designed to facilitate collaborative research, co-designing indigenous protocols, and including under-represented, culturally and linguistically diverse groups? What are effective ways of engaging with these communities, and how can we use generative AI to communicate scientific assessments to different communities [Lansbury et al., 2023]?

Characterizing the literature If the input to the synthesis process diversifies, what is the process of search? How, using which credibility criteria, should we determine corpora of scientific and “non-traditional” research outputs (e.g., indigenous knowledge), grey literature (e.g., blogs, social media, government reports), and select sources to include?

The labeling bottleneck Traditionally, assessment reports relied on domain experts who annotated, often in teams, documents for relevance and for specific aspects of climate impact as part of the evidence gathering and synthesis process. The rapid growth of scientific literature has made this increasingly challenging. Can evidence gathering and screening be automated with recent advances in the use of LLMs for relevance labeling [Thomas et al., 2024]? Can information extraction tasks be off-loaded to LLMs [Mallick et al., 2024]? And to which degree can LLMs be used to compare and synthesize extracted evidence [Joe et al., 2024]?

Trustworthy synthesis IPCC’s assessment reports inform policy-makers. They provide a scientific basis for governments at all levels to develop climate related policies. As a consequence, the reports, and the processes and systems underlying them, need to be trustworthy especially as they become deep learning- or LLM-based [Wang et al., 2024]. From a technological point of view, a process or technology can be trustworthy for *intrinsic* reasons, e.g., because a sufficient level of transparency of the process and technology are provided. Or it can be trustworthy for *extrinsic* reasons, i.e., because we have ways of probing it for properties such as accuracy, reliability, repeatability, resilience, and safety. How can these intrinsic and extrinsic approaches be developed for large-scale evidence synthesis?

Living evidence Evidence changes over time. What are effective update protocols? The temporal dimension plays a big part in evidence outcomes: when should a review be updated? What would the requirements be for self-updating synthesis systems? How should we preserve previous versions [Simmonds et al., 2017]?

Human in the loop synthesis How can we measure the cost, and benefit, of having a human in the evidence synthesis loop [Thomas et al., 2017]? How should we provide feedback to the systems and who provides this feedback? And vice versa, how do we transfer tasks from systems

to people, or trigger interventions from people in complex evidence synthesis tasks? Can interactive or collaborative frameworks from other tasks be adopted [e.g., [Zhang et al., 2021](#)]?

3.3 Obstacles and Risks

We currently know very little about the extent to which automation of key evidence synthesis steps such as query automation, screening automation, and results automation are effective for evidence synthesis in the climate impact domain, and what little we know has usually been learned on English language scientific publications. An important obstacle is the lack of data and benchmarks (especially indigenous knowledge, and low-resource languages) and lack of training corpus for these languages. An important risk is limited success in community engagement, across disciplines and across different types of stakeholder. It is important to do this in the right way by adhering to responsible practices, even though they may not have been established for all types of (non-traditional) domain experts and underrepresented groups [see, e.g., [Lewis et al., 2020](#)].

4 Resources

Progress in the development of effective methods to automate all steps in systematic reviews has been greatly facilitated by the availability of shared resources. In evidence synthesis in healthcare and medicine, for instance, we have witnessed an explosion of methods due to shared tasks and datasets. Prominent examples are the CLEF Technology Assisted Reviews task [[CLEF, 2022](#)] and the CSMed: Meta-dataset for systematic review automation evaluation [[Kusa et al., 2023](#)]. In information retrieval for climate impact there is an urgent need for similar standardized datasets and test collections.

4.1 Proposed Research

Support for the creation of shared resources (data, code, evaluation) requires an understanding of the information and (automated) steps that are needed to accomplish the assessment goals, of the (inter)actions of evidence reviewers working towards the goals, providing broader context as data is gathered and analyzed, dealing with information, and compensating for data quality and integrity issues that may arise during synthesis.

For inspiration, it is instructive to revisit the main steps in the systematic review process [[Cooper et al., 2018](#); [Scells, 2021](#)] – *querying*, *screening*, and *aggregating* – and see what progress has been enabled using shared resources and large language models in recent years. The *querying stage* typically involves steps such as query formulation, query refinement, and query representation. As an example of recent progress using LLMs and building shared resources in the context of querying, [Wang et al. \[2023\]](#) explore the use of LLMs to follow instructions and generate queries with high-precision, while trading this off for recall. Using CLEF resources for evaluation, the authors find that LLMs are a valuable tool for rapid reviews where time is a constraint and trading off higher precision for lower recall is acceptable. The *screening stage* typically involves screening prioritization, cut-off prediction, and document classification. As another example of recent

progress facilitated by the CLEF e-Health,¹ TREC Total Recall,² and TREC Legal³ resources, Stevenson and Bin-Hezam [2023] propose a stopping method based on point processes and test the robustness of a wide range of statistical and neural stopping methods in a series of contrastive experiments. Finally, the *aggregating* stage typically involves information extraction, result synthesis, statistical analyses, and dissemination. As an example of recent progress that uses LLMs and builds on shared resources, Shaib et al. [2023] examine the potential of LLMs to synthesize (that is, aggregate the findings presented in) multiple medical articles in a way that accurately reflects the evidence. Using articles indexed in the Trialstreamer database⁴ and meta-analyses from the Cochrane Library,⁵ the authors conclude that human-written summaries in the (bio)medical domain require a level of synthesis that is not yet captured by today’s LLMs.

Do these findings generalize from the medical domain to information retrieval for climate impact and to all aspects of the systematic review workflow? What are the resources needed to facilitate similar advances in information retrieval for climate impact and how do we create them?

4.2 Research Challenges

This general question for resources in support of information retrieval for climate impact can be addressed at two levels – at the level of individual resources and at the aggregate level of multiple resources. First level, at the level of individual resources, we see the following research challenges.

Document collections Several collections relevant to support research into information for climate impact have been used in recent years. White literature collections used include a subset of the Semantic Scholar Open Research Corpus consisting of 600K climate-related articles [Mallick et al., 2024], subsets (abstracts, titles and metadata (no full text)) from the Web of Science Core Collections databases consisting of 375K [Callaghan et al., 2020] and 565K [Sietsma et al., 2021] articles. Grey literature of different sorts has also been used in the context of climate science. Bai et al. [2024] share a collection of 60K (multi-modal) tweets covering different climate change stances. The GDELT climate news narrative datasets, covering 2009 to 2020, represent television, language, the global perspective and context dating back from half a decade to a decade in all.⁶ How can we create a large-scale, heterogeneous corpus for climate impact, a corpus, moreover that would support a heterogeneous benchmark containing diverse climate-related tasks (like the BEIR dataset in ad-hoc retrieval [Thakur et al., 2021])? How can we create benchmarks, both for static and tracking tasks?

Knowledge sources Knowledge sources play an important role in making sense of both white and grey climate literature. Example knowledge sources used to understand climate science literature at scale include KnowWhereGraph, a geo-knowledge graph that is based on existing standards

¹<https://github.com/CLEF-TAR>

²<http://plg.uwaterloo.ca/~gvcormac/total-recall/>

³<https://trec-legal.umiacs.umd.edu/>

⁴<https://trialstreamer.ieai.robotreviewer.net>

⁵<https://www.cochranelibrary.com/>

⁶<https://blog.gdeltproject.org/four-massive-datasets-charting-the-global-climate-change-news-narrative-2009-2020/>

like RDF, OWL and GeoSPARQL, incorporates custom ontologies, and uses a hierarchical grid for spatial representations; it is built upon many open-source spatial datasets, including an expert knowledge graph built using scientific publication/papers on disaster relief.⁷ Two other important knowledge sources for anomaly and unusual event detection are the iNaturalist species dataset⁸ and the LEO Network.⁹ How can we use these resources to enrich white and grey literature at scale and enable different types of semantic search [Balog, 2018]?

Tools and technology Various domain-specific resources and tools have been developed for information retrieval for climate impact recently. For example, WildfireGPT is an LLM designed to support the analysis of wildfire data, by a diverse set of end users, including researchers and engineers. WildfireGPT is an interesting example of a domain-specific resource not just because of its use cases but also because it uses a broad range of domain-specific data [Xie et al., 2024]. Thulke et al. [2024] introduce ClimateGPT, a family of large language models designed for understanding and responding to information about climate change; the paper demonstrates the effectiveness of adapting a strong general-purpose LLM through continued pre-training on a curated climate corpus and fine-tuning it with high-quality, human-expert-involved instruction data. Mallick et al. [2024] describe an end-to-end decision-support tool that accurately identifies location information within textual documents, enabling extraction of geographic-specific climate trends and topics, to provide a concise understanding of geographic-specific climate change trends. How can tools and technology with such far-reaching potential be made accessible to everyone, not just the technologically privileged?

Rajapakse et al. [2024] describe a widely used library designed to simplify the training, evaluation, and usage of transformer models that operationalizes the idea of “open-source for all,” with (i) a focus on shared innovation, where the knowledge and technology being created and examined is shared and owned by multiple (types of) stakeholders – academic, industrial, governmental, and societal; (ii) a focus on empowering people, equipping them with new tools, rather than replacing them; and (iii) a focus on getting many voices around the table, as the technology and its usage in climate impact is likely to affect many [Shneiderman, 2022]. Can we put this perspective to work for all tools and technology (or technology components) that help make sense of information related to climate impact?

There are multiple research challenges that concern the development or usage at different levels of aggregation.

Transfer & generalization The first challenge concerns the use of the broad range of advances and resources realized in language technology in recent years. How can we transfer these advances to the domain and tasks of information retrieval for climate impact with minimal effort *and* reliable performance? Can we take the findings from evidence synthesis in technology assisted reviews in the medical or legal domain and transfer these to the climate domain? Can these questions be turned into few-shot learning problems [Wang et al., 2025]?

⁷<https://knowwheragraph.org/graph/>

⁸https://www.inaturalist.org/observations?place_id=any&subview=map

⁹<https://www.leonetnetwork.org/en/docs/about/about>

Benchmarking The lack of standardized benchmarks and datasets for validating and comparing information retrieval models for climate impact poses significant challenges to the development, assessment, and comparison of information retrieval technologies in this field. Can we use (very) limited human labeling and rely on LLM-generated labels where large quantities of labels are needed [Thomas et al., 2024] for test collection creation? To which degree would a weak-to-strong alignment perspective [Lyu et al., 2025] be helpful for creating the required benchmarking resources?

Stacking up Recent years have witnessed great progress in the performance of individual components that make up a systematic review workflow – in querying, screening, and aggregating. How do these advances stack up? How do issues such as bias, reproducibility, and performance emerge in complex retrieval for impact pipelines [Hocking et al., 2023]? How can the full review pipeline be optimized effectively? Is the field ready to explore end-to-end learnable approaches to information retrieval for climate impact such as generative information retrieval [Tang et al., 2024]?

Scaling up Managing and processing the vast amounts of heterogeneous data from diverse sources such as sensors, climate models, literature and satellites data poses significant challenges. In the medical domain, organizations such as Cochrane and the Joanna Briggs Institute offer extensive methodological guidance for dealing with large-scale collections, putting a large emphasis on the use of software such as Covidence¹⁰ or JBI Sumari.¹¹ How do such tools scale up to the size and heterogeneity of materials related to climate impact? Furthermore, linking text data from white and grey literature (including news and social media) that describes a climate event to relevant physical measurements about the event, like temperature, rainfall, audio-visual recordings, etc. at a world-wide scale is a challenge that goes far beyond the entity linking challenges that the information retrieval community addresses today [Meij et al., 2013].

4.3 Obstacles and Risks

Collaboration among researchers in information retrieval, natural language processing, data engineering, and climate science, developers, and methodologists is essential for advancing the field of information retrieval for climate impact. The fact that these communities and sub-communities – with different sharing and evaluation cultures – are largely disconnected, is a risk that may hinder progress in the development and uptake of resources and benchmarks.

5 Usage

Recall the core steps from IPCC’s overall evidence synthesis process – querying, screening, and aggregating. In each of these steps it employs provenance tracking as a core component of the workflow. By documenting the origin and reliability of the evidence, WGII provides a foundation for informed decision-making on climate change impacts, adaptation, and vulnerability.

¹⁰<https://www.covidence.org>

¹¹<https://sumari.jbi.global>

5.1 Proposed Research

IPCC’s way-of-working (with a strong emphasis on provenance tracking) comes with several key challenges. First of all, the exponential growth and increasing complexity of scientific literature complicates the IPCC’s mandate to conduct “comprehensive, objective, open and transparent” assessment in line with the IPCC principles, an issue that is exacerbated by the workload required from scientists to engage with assessment [De-Gol et al., 2023; Minx et al., 2017]. Second, the length of the IPCC reports poses significant challenges to access the detailed underlying evidence in the main reports. For example, in the most recent assessment round (AR6), the two main reports exceed 2,000 pages each, with one exceeding 3,000 pages. Including additional specialized reports and summaries, a total of $\sim 12,000$ pages was reached. The sheer volume hinders the effective dissemination of important scientific findings that do not feature in the summaries.

Assuming that the community manages to make progress along the lines suggested in Section 2, 3, and 4, how and where can the resulting advances be made to work for IPCC’s assessments, given the challenges outlined above? In particular, how can we navigate the vast and ever-expanding corpus of white and grey literature on climate in an efficient manner to enable comprehensive and credible consensus building processes? How to use such tools to effectively communicate the extensive findings in the lengthy IPCC reports while maintaining the integrity of the assessments? How to determine the weight of heterogeneous evidence, complexity, uncertainty and the level of confidence?

5.2 Research Challenges

Al Khourdajie [2024] summarize several current lines of work to structure the assessment work, in a way that addresses the questions listed above and that provides connecting points for the agendas in Section 2, 3 and 4. The IPCC scoping process could use *evidence maps* to highlight key topics. Once outlines are agreed, *topic discovery* could help authors of assessment reports plan drafts by clustering publications and topics. A *hybrid exploration* approach could then be used to inform assessment drafting. *Collaborative platforms* could then be used to expand engagement, while *exploration* tools would help to expand coverage.

Synthesis and evidence maps An evidence map is a visual representation of the available research on a particular topic. It provides an overview of the extent, range, and nature of evidence, highlighting where research exists and, importantly, where gaps remain [Hempel et al., 2014]. How can LLMs be used to identify and classify white and grey literature publications and provide rapid and comprehensive coverage?

Topic and event discovery How can traditional information retrieval and LLM-based methods be used to identify and track topics and events in white and grey literature so as to reveal gaps?

Hybrid exploration While expert judgment remains crucial for assessing literature that is meant to help inform policy-makers, how can we organize effective iterative human-in-the-loop (or machine-in-the-loop) approaches that alternate between query reformulation and retrieval steps and are likely to identify important thematic clusters and yield content-based insights?

Collaborative platforms De-Gol et al. [2023] describe their experience with a collaborative technology platform, tailored to support an updated process of elaborating IPCC reports. They argue that it could greatly enhance IPCC assessments by making them more open and accessible, further increasing transparency. What sort of content-based recommendation methods, based on ideas in Sections 3 and 4, could help to create engagement?

Exploration How can modern information retrieval methods, in combination with visualization and semantic tools, support the discovery of publication networks, enable swift navigation and simplify subsequent literature synthesis?

5.3 Obstacles and Risks

Beyond the data, technological, and methodological challenges listed above, the uptake of outcomes from the research suggested in Section 2, 3, and 4 in climate impact assessments faces ethical and social challenges [Ghosh et al., 2022]. Ensuring that the increased use of advanced information retrieval technology promotes equity and justice is essential – often, the people most affected by climate change, are the least involved in the creation of the technologies, or the assessments of climate change. Avoiding the creation of new inequalities or exacerbating existing ones is a key consideration. Ensuring transparency and accountability in the use of new information retrieval technologies is crucial for building trust and legitimacy. This includes clearly documenting the methods, data, and assumptions used in the assessment. Finally, engaging stakeholders in the development and use of new information retrieval technologies is essential for ensuring that assessments are relevant and useful. This requires effective communication and collaboration.

6 Perspective

Climate change is real. The world has failed to avoid crossing the 1.5° C threshold set by governments to avert the worst impacts of global warming. This is a new reality. However, “[f]or the time being, it’s just one metric and one year, but researchers say that it nonetheless serves as a stark reminder that the world is moving into dangerous territory — perhaps more quickly than previously thought” [Tollefson, 2025]. The information retrieval community has a unique opportunity to inform and foster a community of researchers and practitioners that contribute to effective information retrieval solutions in support of research and decision-making to help address the reality of climate impact [Sietsma et al., 2024]. The agenda setting activities of this workshop were meant to do just that.

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B Process

This paper grew out of the half-day SIGIR 2024 workshop on Information Retrieval for Climate Impact [van den Hurk et al., 2024], which was held at SIGIR 2024 on July 18, 2024. The workshop was organized by BvdH, MdR, FS, who invited a number of researchers from the information retrieval and climate science domains to form the program committee. The workshop itself consisted of two parts. The first part was focused on gaining a better understanding of the information retrieval challenges faced by IPCC WGII, with four presentations on the topics underlying the four

core sections of this paper (Sections 2–5) as well as eight presentations that were selected based on an open call for contributions. The second part of the workshop consisted of a collaborative writing session, with four groups working on the topics underlying the four core sections of the paper, using “seed topics” contributed by members of the program committee. The workshop organizers used the presentations and the outputs of the writing session to draft a first version of the paper, which was then shared with the entire author team for review and editing.

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